

From Graphical Models to Generative AI

Chapter 2: From Local Factors to a Global Distribution

Michael (Misha) Chertkov

MATH 577-001 University of Arizona

Lecture 2 — Book + Notebook as a Computational Laboratory

Wednesday, August 26, 2026 Mathematics 514

Goal today: learn the mathematical object *and* how we will use the living book.

How we will use the living book

Book $\longrightarrow$ **Whiteboard** $\longrightarrow$ **Notebook** $\longrightarrow$ **Exercise / Problem / Challenge**

Define

notation, assumptions,
exact statements

Rebuild

derive the object and
understand the
structure

Compute

verify, visualize,
reproduce

Push

perturb, stress-test,
correct, extend

The notebook is not a code listing to memorize. It is an executable mathematical laboratory.

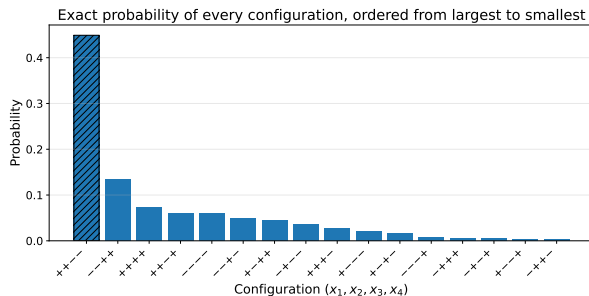
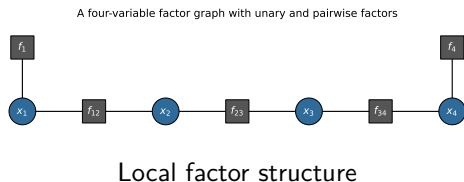
Chapter 2 starts with one exact object

$$w(\mathbf{x}) = \prod_{a \in \mathcal{F}} f_a(\mathbf{x}_a), \quad Z = \sum_{\mathbf{x}} w(\mathbf{x}), \quad p(\mathbf{x}) = \frac{w(\mathbf{x})}{Z}.$$

Representation	Which variables enter which local factors?
Normalization	What is Z ?
Inference	What are marginals and expectations?
Optimization	Which configuration is MAP?
Learning	How do parameters/factors connect to data?
Generation	How do we produce samples from p ?

For four binary variables: exact enumeration is our ground truth.

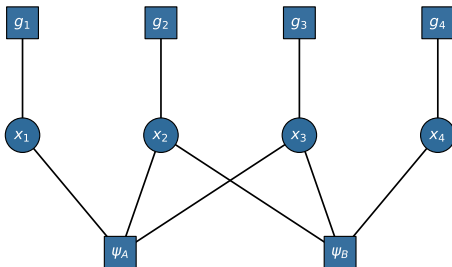
Notebook 1: an exact sandbox before approximation



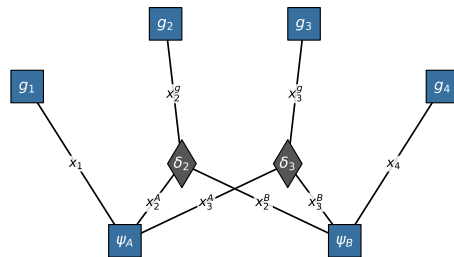
ch02/ch02_01_factorization_and_enumeration.ipynb

Same probability law, different local representation

Ordinary factor graph



Equivalent normal factor graph



$$Z_{\text{FG}} = Z_{\text{NFG}}$$

Duplicate local variables are harmless only when equality factors enforce consistency.

From an example to an Exercise / Problem / Challenge

1. Read the definition / derivation.
2. **Predict** before running the notebook.
3. Reproduce the exact baseline.
4. Change one factor, parameter, constraint, or representation.
5. Explain what changes — and what must remain invariant.
6. Push until an assumption, approximation, or representation breaks.

Portfolio Window 1

Exercise (E) = 1 credit

Problem (P) = 2 credits

Challenge (C) = 3 credits

Due: **Fri., Sept. 18, 6:00 PM**

Suggested filenames:

W1-E-short-topic.pdf

W1-P-short-topic.pdf

W1-C-short-topic.pdf

Three things to carry forward

1. **Local structure builds a global distribution.**
2. **Exact small models are not toys: they are references for later approximations.**
3. **Representation matters computationally, even when the represented law is unchanged.**

For every notebook: first understand the object, then reproduce it, then change something.

Next: conditional independence and more graphical representations.